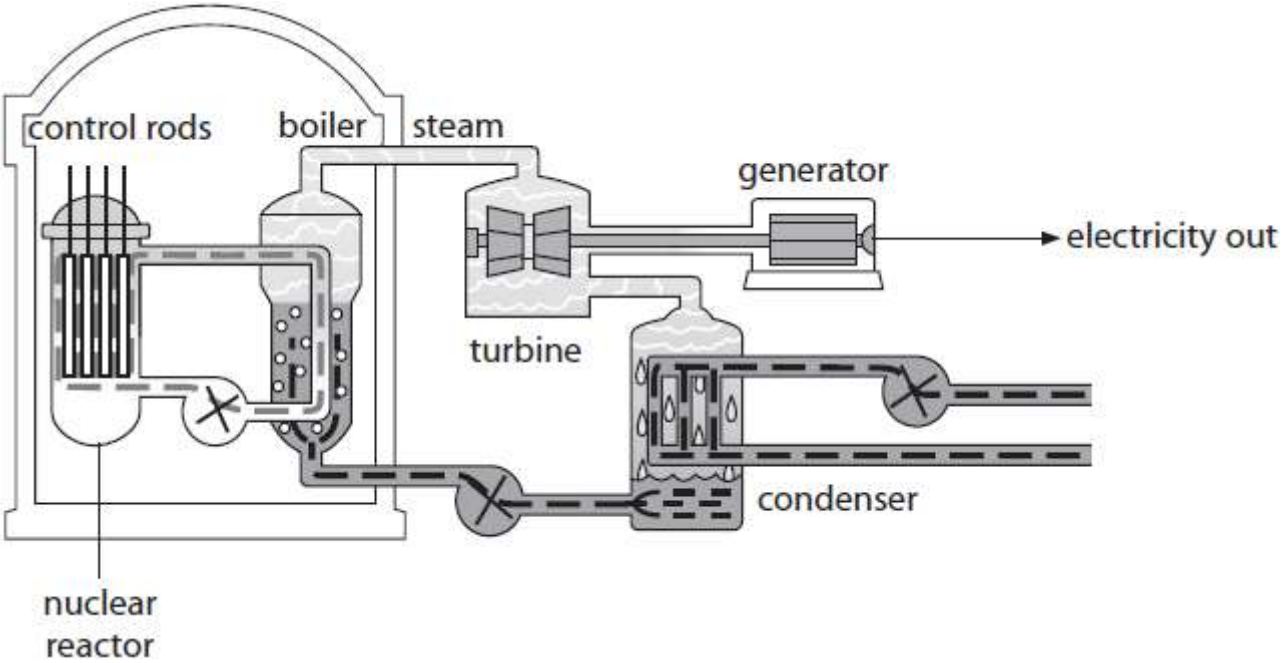
Questions

Q1.

The diagram shows parts of a nuclear power station.



(i) Which part of the power station provides thermal (heat) energy from a chain reaction?

Put a cross (☒) in the box next to your answer.

(1)

☐ **A** nuclear reactor

☐ **B** turbine

☐ **C** generator

☐ **D** condenser

(ii) Which part of the power station transfers kinetic energy into electrical energy?

Put a cross (☒) in the box next to your answer.

(1)

☐ **A** nuclear reactor

☐ **B** boiler

☐ **C** turbine

☐ **D** generator

Q2.

(a) The nucleus of a hydrogen atom can be represented by this symbol:

$$^{1}_{1}H$$

(i) Complete this sentence by putting a cross (☒) in the box next to your answer.
The symbol shows that the nucleus consists of

(1)

☐ **A** 1 proton and 1 neutron
☐ **B** 1 proton only
☐ **C** 1 neutron only
☐ **D** 1 neutron and 1 electron.

(ii) Two other isotopes of hydrogen are deuterium (D) and tritium (T).
Their nuclei can be represented by these symbols:

$$^{2}_{1}D \qquad\qquad ^{3}_{1}T$$

State how these symbols show that they are isotopes of hydrogen.

(1)

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(b) Nuclear fusion can occur if a deuterium and a tritium nucleus can be brought close enough to each other.

This fusion produces a helium nucleus and releases a neutron.
(i) Compare the charges of a helium nucleus and a neutron.

(2)

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(ii) Nuclear fusion is an important process.
Scientists have said *'without nuclear fusion, there would be no life on Earth'*.
Explain why nuclear fusion is important to life on Earth.

(2)

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*(c) Neutrons are also released during nuclear fission.
Describe how the neutrons released in nuclear fission are used to produce a controlled chain reaction in a nuclear reactor.

(6)

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(Total for Question = 12 marks)

Q3.

The fuel in a nuclear power station is an isotope of uranium.
Nuclear fission is the reaction that happens in a nuclear power station.
Explain what happens when nuclear fission occurs.

(2)

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Q4.

Use words from the box to complete the following sentence.

The words may be used once, more than once, or not at all.

(2)

alpha	atom	beta
molecule	neutron	nucleus

During nuclear fission, a uranium-235 ..... splits

when it absorbs a slow moving ..... .

Q5.

Answer the question with a cross in the box you think is correct ☒ . If you change your mind about an answer, put a line through the box ☒ and then mark your new answer with a cross ☒ .

This question is about nuclear reactions.

When a uranium-235 (U-235) nucleus absorbs a neutron, the nucleus splits into smaller parts.

This reaction is called nuclear fission.

How many daughter nuclei are produced from the fission of one nucleus of uranium-235 (U-235)?

(1)

☐ **A** two

☐ **B** three

☐ **C** four

☐ **D** six

(Total for question = 1 mark)

Q6.

This question is about nuclear reactions.

(i) When a uranium-235 (U-235) nucleus splits, neutrons are also emitted.

The neutrons may start a chain reaction.
Describe what is meant by a chain reaction.
You may draw a diagram to help with your answer.

(2)

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(ii) In the nuclear reactor of a power station, the chain reaction has to be controlled.

Explain the action of a moderator in a nuclear reactor.

(2)

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(iii) In a nuclear reactor there are 2.0×10^{17} fission reactions each second.

Each fission reaction releases 4.0×10^{-11} J of energy.
Calculate the energy released in 1 second.

(2)

energy released in 1 second = ... J

(Total for question = 6 marks)

Q7.

(i) State the name of an instrument that can be used to measure radioactivity.

(1)

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(ii) State **two** sources of background radiation.

(2)

1 .....

2 .....

(Total for question = 3 marks)

Q8.

Figure 17 shows a Geiger-Müller (GM) tube used for measuring radioactivity.

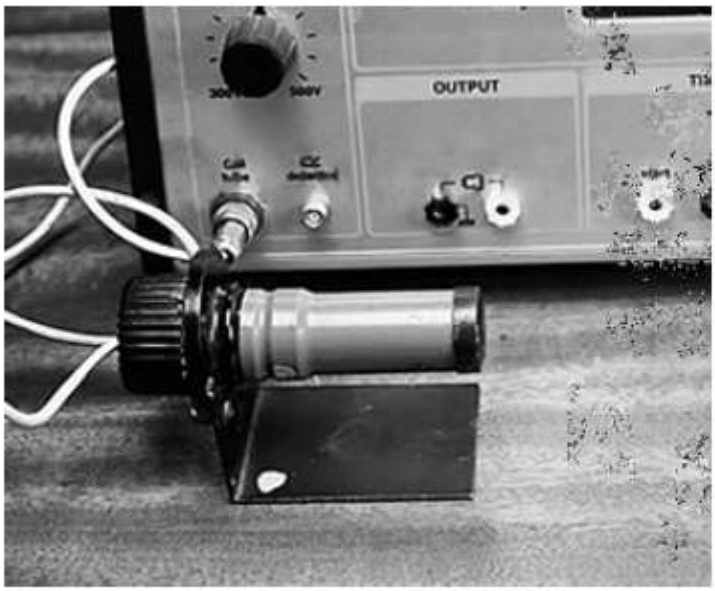

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Figure 17

Describe how a teacher should use a Geiger-Müller (GM) tube to compare the count-rates from two different radioactive rocks.

(4)

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(Total for question = 4 marks)

Q9.

Everyone is exposed to background radiation. Some of this radiation comes from natural sources.
(a) (i) One example of a source of background radiation that does not occur naturally is radiotherapy.
State **one** other source of background radiation that does not occur naturally.

(1)

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(ii) Radon gas is a natural source of background radiation.

In some parts of the country, a lot of the background radiation comes from radon gas.

Explain why there is no radon gas in some other parts of the country.

(2)

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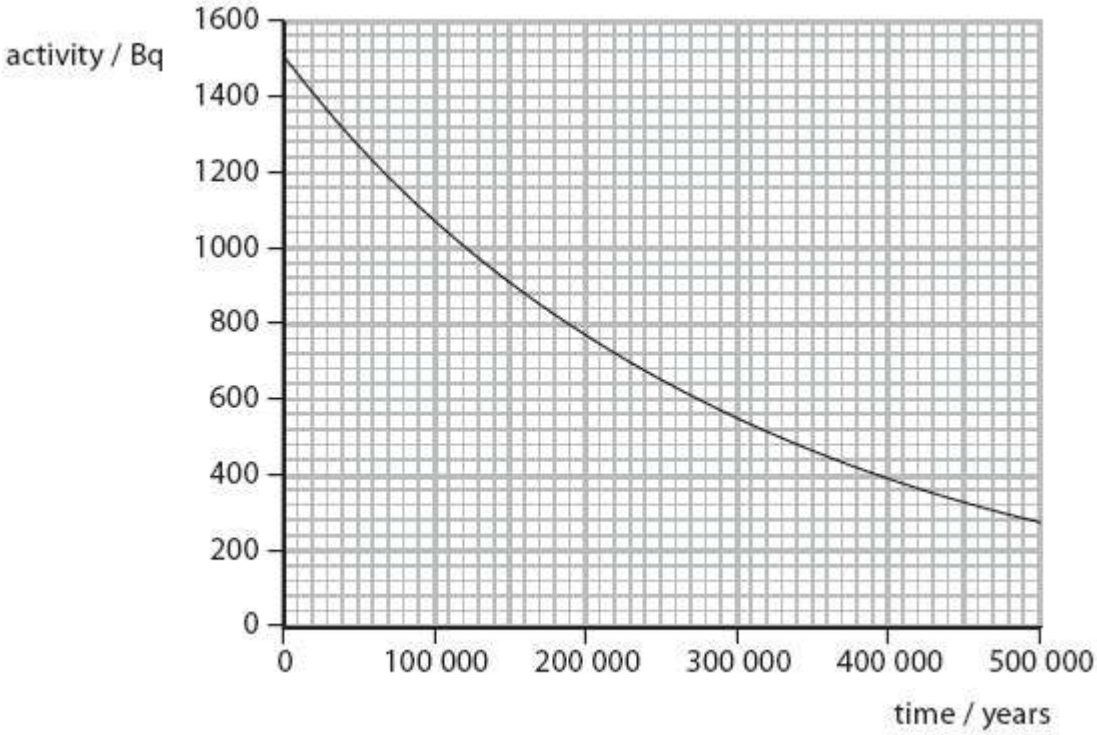
(b) Technetium-99 is one of the radioactive isotopes in nuclear waste.
The graph shows the decay curve for technetium-99.



(i) Use the graph to show that the half-life of technetium-99 is about 200 000 years.

(2)

(ii) Technetium-99 emits beta particles.

Give **one** reason that beta particles can cause harm to people.

(1)

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.....

*(c) There are many radioactive isotopes in nuclear waste.
Technetium-99 is just one of these isotopes.

People are worried about how we should deal with nuclear waste.

Explain why it is difficult to deal with nuclear waste safely.

(6)

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(Total for Question is 12 marks)

Q10.

Explain **one** precaution that is taken in hospitals to limit the risks of exposure to radiation.

(2)

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Mark Scheme

Q1.

Question Number	Answer	Acceptable answers	Mark
(i)	**A** nuclear reactor		(1)

Question Number	Answer	Acceptable answers	Mark
(ii)	**D** generator		(1)

Q2.

Question Number	Answer	Acceptable answers	Mark
(ai)	**B** 1 proton only (1)		(1)

Question Number	Answer	Acceptable answers	Mark
(aii)	Same number of protons (as hydrogen) or same atomic number(as hydrogen) (1)	Same proton number(as hydrogen) / (they all) have one proton / (their) proton number is 1 accept bottom number is 1/the same NOT same mass / nucleon number NOT same atomic mass ignore references to electrons / neutrons	(1)

Question Number	Answer	Acceptable answers	Mark
(b)(i)	Helium (nucleus has) positive/+ (charge) (1) Neutron has no/zero/0 (charge) (1)	helium is +(any number >0 and <5) helium has a larger/bigger charge neutron is neutral /neutrally charged/uncharged ignore references to nuclear fusion or masses	(2)

Question Number	Answer	Acceptable answers	Mark
(b)(ii)	An explanation linking (Nuclear fusion/it) occurs in the Sun (1) (The Sun / Fusion provides) energy/heat/light (needed for life on Earth) (1)	 (nuclear fusion/it) is the Sun's energy source OR (it) occurs in stars any valid use of fusion in Sun or stars e.g. • without heat (from Sun) Earth would freeze/have no life • new/heavier elements are made (by fusion/ in stars) eg creates helium	**(2)**

Question Number		Indicative Content	Mark
QWC	* **(c)**	A description including some of the following points Stages involved in a chain reaction: o (neutrons released go on to) collide with other nuclei o causes nuclei to become unstable o (nuclei) split/fission (into daughter nuclei) o releases **more** neutrons o releases energy Control: -Action of the moderator o neutrons need to be slowed down/turned into thermal neutrons o to increase chance of collision o this is achieved with a moderator o carbon/graphite/water/heavy water can be used -Action of control rods o number of neutrons available for collision needs to be controlled o so that reaction proceeds at a steady rate / does not increase o this is achieved by control rods absorbing neutrons o boron / silver/indium/cadmium can be used. Many candidates repeat parts of the question Do NOT give credit for these statements eg neutrons are released during fission	**(6)**

Level	0	No rewardable content
1	1 - 2	• A limited description which gives one relevant fact e.g. (neutrons) cause atoms to split. OR (during fission of uranium atom) neutrons collide with atoms OR (nuclear fission) releases energy OR (3) neutrons are released and two of them are absorbed/taken away • the answer communicates ideas using simple language and uses limited scientific terminology • spelling, punctuation and grammar are used with limited accuracy
2	3 - 4	• A simple description, giving more than one fact, about a chain reaction or control OR at least one fact about both. e.g. a neutron collides with (uranium) atoms and causes them to split (into daughter nuclei) OR atoms split releasing **more** neutrons OR an atom splits and releases energy OR (neutrons) cause atoms to split and there are (control) rods to control the neutrons. OR control rods can be lowered into the reactor to absorb neutrons • the answer communicates ideas showing some evidence of clarity and organisation and uses scientific terminology appropriately • spelling, punctuation and grammar are used with some accuracy
3	5 - 6	A detailed description involving:- • more than two stages of the chain reaction • OR a description involving more than one stage of the chain reaction AND at least one detail about control. • OR a description involving more than one detail about control AND at least one detail about the chain reaction. e.g. Neutrons are slowed down by graphite/water. This makes them more likely to collide with other nuclei. OR neutrons collide with other nuclei and cause them to split releasing more neutrons AND these neutrons hit another nuclei causing it to split OR neutrons collide with other nuclei and cause them to split releasing more neutrons AND there are (control) rods to control the neutrons OR neutrons collide with uranium nuclei causing them to split and release more neutrons. Control rods of boron absorb some of the neutrons. • the answer communicates ideas clearly and coherently uses a range of scientific terminology accurately. uses nuclei split and not atoms split. • spelling, punctuation and grammar are used with few errors.

(Total for Question =12 marks)

Q3.

	Answer	Acceptable answers	Mark
	An explanation linking any two of the following points • a neutron(s)(1) • hits nucleus/nuclei (1) • uranium/nucleus splits (1) • (producing) neutrons /daughter nuclei/ energy / Kr and Ba (1)	collides/is absorbed breaks/divides ac cept chain reaction for 1 mark if no other mark awarded accept a correctly labelled diagram	**(2)**

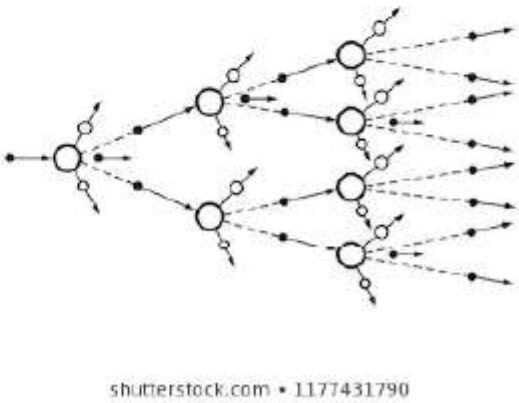
Q4.

Question Number	Answer	Acceptable answers	Mark
	nucleus (1) neutron (1)	**Answers in this order only**	**(2)**

Q5.

Question number	Answer	Additional guidance	Mark
	A two B is not correct as a uranium nucleus does not split to give 3 daughter nuclei C is not correct as a uranium nucleus does not split to give 4 daughter nuclei D is not correct as a uranium nucleus does not split to give 5 daughter nuclei		**(1)** **AO1**

Q6.

Question number	Answer	Additional guidance	Mark
i	A description including: EITHER neutrons are absorbed by uranium nuclei/atoms (1) more neutrons are produced/emitted (which are absorbed by uranium nuclei) (1) OR diagram (no labels needed) two nuclei/atoms splitting (1) four or more nuclei/atoms splitting (1)  	ignore any reference to bonds accept hit /collide with accept a controlled chain reaction diagram for 2 marks	(2) AO1

Question number	Answer	Additional guidance	Mark
ii	An explanation linking: (because the moderator/it) slows down/increases the chance of absorption of(1) neutrons (1)		(2) AO1

Question number	Answer	Additional guidance	Mark
iii	substitution(1) $2(.0 \times 10^{(17)}) \times 4(.0 \times 10^{(-11)})$ evaluation (1) $8(.0) \times 10^{6}$ (J)	accept 8000000(J) accept 8MJ 8 to any other power of ten scores 1 mark award full marks for correct answer without working.	**(2)** **AO2**

Q7.

Question Number	Answer	Additional guidance	Mark
(i)	Geiger (Müller counter) (1)	GM {tube/meter} or other appropriate detector e.g. dosimeter, film badge, scintillation counter accept incorrect spellings such as "giga" ignore radioactive counter	**(1)**

Question Number	Answer	Additional guidance	Mark
(ii)	any **two** acceptable sources from :		(2)
	cosmic (rays) (1)	cosmic microwave background radiation (CMBR)	
	Sun (1)		
	rocks / ground (1)		
	{nuclear / atomic} tests / nuclear waste (1)	accept nuclear accidents (Chernobyl, Fukushima etc)	
	(nuclear) power stations (1)		
	plant (sources) (1)		
	buildings (1)		
	food (1)	accept named foods	
	water (1)		
	medical (1)	accept X-rays, radiotherapy	
	radon (1)	ignore alpha, beta, gamma	

Q8.

Question Number	Answer	Additional guidance	Mark
	a description to include: 1. put rock(s) in front of/near tube (1) 2. measure (count rate) separately for the two different rocks (1) 3. measure each count for the same time period (1) 4. keep source-detector distance the same for both rocks (1) 5. take (into account)/measure background count (1) 6. repeat readings and take average(s) (1)	not 'in' tube keep rocks apart	**(4)** AO 2 2

Q9.

	Answer	Acceptable answers	Mark
(a)(i)	any one of X-ray (machines) / smoke alarms/ nuclear/ radioactive waste (1)	nuclear weapons (tests) nuclear power plants (medical) tracers/technetium	**(1)**
(a)(ii)	an explanation linking: comes from granite / rocks (1) none/ less of these (rocks) in some areas (1)	in some areas/Cornwall/Aberdeen the second mark is dependent on the first.	**(2)**
(b)(i)	suitable lines on graph to show halving after about 200 000 years (2) • horizontal line at 750 +or –50 Bq on y-axis to curve (1) • meeting (by eye) vertical line from	use of data from graph to show halving after about 200 000 years 1500/2 =750(Bq) or 1600/2=800(Bq) gives a half-life of 210,000 +or- 20 000 (years)	**(2)**

	x-axis between 190,000 years and 230,000 years (1)		
(b)(ii)	any one of • penetrates/passes through the skin (1) • ionises (1) • damages tissue/ cells/DNA (1) • mutates cells/DNA(1) • causes cancer(1)		**(1)**

Total marks for question = 12

		Indicative Content	**Mark**
QWC	***(c)**	an explanation which may include some of the following points: properties of nuclear waste radioactivity is dangerous some isotopes in nuclear waste have long half-lives/radioactive for thousands of years products of fission are warm identified radiation from nuclear waste e.g alpha, beta, gamma problems caused by nuclear waste leakage of radioactivity contamination of ground/sea water/lakes /rivers contamination of crops/fish/animals/drinking water harm to humans/cancer/radiation poisoning/ damage to cells/mutation of cells or DNA difficulty in transporting safely/ stolen by terrorists fears of local people solutions for dealing	**(6)**

		with nuclear waste safely long term storage, underground /under the sea radiation shielding, lead/steel/concrete/ containers, sealed in glass. human safety, radiation suits, using tongs/lead jackets safe location, away from people/remote areas/sea cooling, ponds information to persuade local people of safety	
Level	**0**	No rewardable content	
1	**1 - 2**	• a limited explanation mentioning at least one point, but without linking, e.g. radioactivity is dangerous ; nuclear waste should be stored underground ; terrorists might steal nuclear waste; • the answer communicates ideas using simple language and uses limited scientific terminology • spelling, punctuation and grammar are used with limited accuracy	
2	**3 - 4**	• a simple explanation mentioning two points with an appropriate linkage e.g. nuclear waste is dangerous and it must be stored underground ; the isotopes in nuclear waste have long half-lives so they must be stored for a long time; • the answer communicates ideas showing some evidence of clarity and organisation and uses scientific terminology appropriately • spelling, punctuation and grammar are used with some accuracy	
3	**5 - 6**	• a detailed explanation mentioning a range of points with appropriate linkages e.g. gamma rays from nuclear waste causes damage to cells so it must be stored away from where people live ; the isotopes in nuclear waste have long half-lives so they must be stored underground or in remote areas; • the answer communicates ideas clearly and coherently uses a range of scientific terminology accurately • spelling, punctuation and grammar are used with few errors	

Q10.

	Answer	Acceptable answers	Mark
	An explanation linking any suitable precaution to a sensible reason: Eg lead/shielding (1) (because it) stops/absorbs radiation (1) Use of radiation meters (1) measure/warn of radiation received (1) put up signs (1) (to)keep people away from radiation (1) increasing distance (1) (to)reduce intensity (1) to reduce dose(1) by limiting the number of X-rays taken (1)	source locked away for shielding (worker) leaves room inverse square law	**(2)**